

## A Written de Sitter–Unruh Modified-Inertia Action (v4)

**The passive-frame constraint structure closes (a machine-verified Dirac system with zero propagating frame modes), the nonlocal operator is rigorously defined by a positive Herglotz–Nevanlinna measure, the two-point theory is causal, ghost-free, Cassini-safe and nonlinearly stable, the Einstein-aether strong-coupling wall is mis-applied, and gravitational lensing is viable but tuned**

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### Abstract

The de Sitter–Unruh reading of the MOND acceleration scale,  $a_0 = cH_\Lambda/Z = c^2\sqrt{(\Lambda/32\pi)} = 9.36 \times 10^{-11} \text{ m s}^{-2}$ , posits that the scale is a modified-**inertia** effect of the cosmic (dark-energy) horizon rather than a modification of gravity. This paper writes the explicit nonlocal modified-inertia action and carries its covariant analysis to a definite boundary, reasoning throughout from the framework’s own premises (a passive frame, with the MOND in the matter sector) rather than the standard aether/MOND lens. Results, each with reproducible verification (both  $a_0$  footings): **(1) the worldline/matter sector is machine-verified** ( $v(y)=\sqrt{(1+1/y)}$  to  $3 \times 10^{-13}$ , Newton, deep-MOND, baryonic Tully–Fisher, the dS-Unruh-forced  $\sqrt{2}$ -DC-weight external-field kernel, exact ghost-freedom). **(2) The modified-inertia realization evades the Cassini quadrupole** by  $\sim 7$  orders (deep-Newtonian  $v-1 \approx 7 \times 10^{-7}$ ), where the modified-gravity realization fails at  $+6$  to  $+14\sigma$ . **(3) Promoting the frame field covariantly, MOND does not force modified gravity** — the matter source is  $l=0$  (soaked by the unit-timelike multiplier), evading the Bianchi shear lock. **(4) The mixed matter–aether two-point propagator is causal and ghost-free in closed form** (retarded construction; Källén–Lehmann spectral positivity across the whole cut; principal symbol = the GR light-cone). **(5, correcting v2) The Einstein-aether strong-coupling wall does not wall the framework**: that wall is a property of a *propagating* aether mode whose kinetic norm vanishes, but the framework’s frame is **passive** (its own theory: an algebraic constitutive law, an SME background, the cosmic rest frame), with zero propagating aether modes — so the wall’s object is undefined for it. The wall binds only the rejected AeST/khronon *modified-gravity* limb. **(6) Gravitational lensing is viable but fine-tuned**: light is massless, so lensing must come from the ghost-condensate dark sector, which reproduces the observed lensing=dynamics relation only if placed by hand to equal the MI dynamical excess (per-galaxy) — not excluded, not predicted. New in v4, addressing three formal completeness questions: **(7) the passive-frame constraint structure is a machine-verified closed Dirac system**: the full curved-spacetime Dirac analysis terminates at the secondary level (the unit-norm sector is genuinely second-class — its  $2 \times 2$  Dirac block has determinant  $4(u \cdot u)^2 \rightarrow 4$  on-shell — with no tertiary tower), leaving **zero propagating frame degrees of freedom** (2 total: the GR graviton). The crux — that  $u$  sits *inside*  $\square_\perp u$  and might thereby acquire a kinetic term — resolves negatively: the exact quadratic symbol at every order is  $S_n = (-1)^n k^2 \perp k_0^{2n}$ , whose only root is  $k_0=0$  independent of spatial  $k$ , i.e. a transport-along- $u$  ODE with **no wave-cone and zero group velocity** — the matter coupling does *not* promote  $u$  to a dynamical field. **(8) The nonlocal operator  $K(\square_\perp u)$  is rigorously defined** by the Borel functional calculus of the (essentially self-adjoint) directional operator  $\square_\perp u$ , with a canonical **Herglotz–Nevanlinna spectral representation** carrying a *unique positive Borel measure* (closed-form density, Källén–Lehmann-positive);  $K$  is bounded ( $\|K\| \leq 1$ , operator-monotone) on all of  $L^2$ , causal-retarded, and its non-entire branch cut is the physical de Sitter–Unruh emission continuum, not a ghost. **Nonlinear** classical stability follows from the same no-wave-cone symbol to all orders (hyperbolicity inherited from GR); **radiative** stability is structurally protected — the  $(\nabla u)^2$  aether counterterm is symmetry-forbidden by

the purely longitudinal  $u$ -derivatives, so  $\alpha_1=\alpha_2=0$  does not run, and  $a_0$  is IR-protected by  $\sqrt{z}$  analyticity plus khronon shift symmetry — but is argued at the power-counting level, not yet by a full covariant one-loop de Sitter computation. **Open:** that covariant one-loop calculation, and a first-principles condensate profile law. The MOND sign remains a **postulate** and  $a_0$ 's value is underived. An effective one-scale completion carried to a sharp, named boundary — **not a finished theory and not a theory of everything**.

### 1. Setting and scope

The framework reads  $a_0$  as the acceleration at which a body's inertial response to the de Sitter vacuum departs from Newtonian, with the Deser–Levin temperature  $T(a) = (\hbar/2\pi k_B c) \sqrt{(a^2 + (cH_\Lambda)^2)}$  a horizon floor and the interpolation  $g_{\text{obs}} = \sqrt{(g_{\text{bar}}^2 + g_{\text{bar}} a_0)}$ ,  $v(y) = \sqrt{(1+1/y)}$ . The earlier theory-of-everything and Standard-Model claims were publicly retracted (2026-06-23) and are not reasserted; this is a one-scale effective-theory construction with named inputs, each labeled **derived**, **postulated**, or **open**. Crucially — and this is the correction to  $v2$  — the framework is *modified inertia*: the MOND lives in the **matter** sector, and the frame field  $u^\mu$  is a **passive**, horizon-anchored reference (the framework's own paper: an *algebraic constitutive law*  $g_{\text{bar}}=G(a)$  with no frame equation of motion; an SME gravity-sector *background*; the Cassini-constrained cosmic rest frame). It is **not** AeST/Einstein-aether, where a dynamical aether *carries* the MOND — the theory that fails Cassini.

### 2. The action

With signature  $(-+++)$ ,  $S[x,g,u] = S_{\text{EH}}[g] + S_u[g,u] + S_{\text{matter}}[x,g,u]$ : host GR unmodified; a unit-timelike frame constraint  $S_u = -\int \sqrt{-g} (\lambda/2)(u^\mu u_\mu + 1)$  (no aether kinetic term); and the modified-inertia content in the matter kinetic sector,

$$S_{\text{matter}} = -\frac{1}{2} \int d^4x \sqrt{-g} \rho_m [s u^\mu K(\Box_u/a_0^2) u_\mu], \quad K(z) = \frac{\sqrt{1+4z}-1}{2\sqrt{z}}, \quad \Box_u f = u^a \nabla_a (u^b \nabla_b f).$$

$K$  is non-polynomial with a branch cut and a single healthy pole (residue +1). The equivalent Galley doubled-worldline form uses the conservative gradient  $G(a)^{\mu=a} \mu_\mu \text{fw}(|a|/a_0)$ ,  $\mu_{\text{fw}}$  the exact inverse of  $v(y)=\sqrt{(1+1/y)}$ ; the external-field kernel  $\theta(y)=\theta_0/(1+(\theta_0-1)y^2)$ ,  $\theta_0=\sqrt{2}$ , has its shape forced by the de Sitter Wightman function. Here  $s=-1$  is the **MOND-sign postulate**,  $a_0=cH_\Lambda/Z=9.36 \times 10^{-11}$  (canonical),  $1.13 \times 10^{-10}$  (alternate footing).

### 3. Verified: the worldline sector and the Cassini evasion

Machine-checked, both footings: the circular-orbit reduction reproduces  $v(y)=\sqrt{(1+1/y)}$  to  $3.35 \times 10^{-13}$ ; Newton, deep-MOND, and baryonic Tully–Fisher  $v^4=GM/a_0$  (exact); the form-factor  $\leftrightarrow$  RAR identity  $g_{\text{obs}}=\sqrt{(g_{\text{bar}}^2+g_{\text{bar}} a_0)}$ ; the  $\sqrt{2}$ -DC-weight external-field kernel; and ghost-freedom (single pole +1 vs the Ostrogradsky ghost of the local truncation). **Cassini:** the anisotropic galactic external field ( $a_{\text{ext}}=2.29 a_0$ ) enters only Saturn's inertial response, deep-Newton-suppressed by  $v-1 \approx 7 \times 10^{-7}$ ; the  $l=2$  quadrupole is  $\sim 7.4 \times 10^{-34} \text{ s}^{-2} \sim 7$  orders below the  $5.2 \times 10^{-27}$  ceiling. The modified-gravity realization instead fails at  $+6$  to  $+14\sigma$ . This is Milgrom's [2009] statement that the inner-solar-system quadrupole is an AQUAL/QUMOND effect absent in modified inertia, realized explicitly.

### 4. The covariant completion, on the framework's own terms

Because the frame is passive and the MOND is in the matter sector, the covariant question is *not* whether a dynamical aether is strongly coupled — it is whether the passive-frame + nonlocal-K matter system is well-posed. Six results:

(a) **MOND does not force modified gravity.** Varying  $S_{\text{matter}}$ , the source it adds to  $u^\mu$ 's field equation is  $J^\mu = -\rho_m s k u^\mu$ :  $l=0$ , parallel to  $u^\mu$ , soaked by the multiplier  $\lambda$ . It never populates the  $l=2$  traceless-

shear channel whose divergence is the Bianchi lock that forces AeST's Cassini quadrupole. So the covariant lift does not collapse to modified gravity; a passive frame carries at most  $Q_2^A u \lesssim 7 \times 10^{-35} \text{ s}^{-2}$  over an open, ghost-free region.

**(b) The mixed two-point propagator is causal and ghost-free in closed form.** The dynamical- $u$  variation resums to  $Q(z)=K+2zK'=2\sqrt{z}/\sqrt{(1+4z)}$ ; the retarded memory kernel has strictly causal support (an advanced control correctly flips it), Titchmarsh UHP-analyticity holds, and Källén–Lehmann spectral positivity  $\rho(s) \geq 0$  across the whole cut is verified in closed form (a control correctly flags a known Ostrogradsky ghost as  $\rho < 0$ ). The branch cut is a healthy radiation continuum, not a ghost.

**(c) The principal symbol is the GR light-cone,**  $D(w)/w = 1 + i a_0/2w - a_0^2/8w^2 + \dots$ , the nonlocal correction  $O(a_0/w)$  UV-subdominant and IR-gapped — hyperbolic, no secular runaway. So the passive-frame theory is **principal-symbol-well-posed**.

**(d) The Einstein-aether strong-coupling wall is mis-applied (correcting v2).** That wall is, by construction, a property of a *propagating* aether/khronon mode whose canonical kinetic norm vanishes ( $M_{sc} \sim \sqrt{N} M_{PI}$ ; the mode speeds carry the vanishing norm in the denominator) [Blas–Pujolàs–Sibiryakov 2009; Gong et al. 2018]. The framework's frame is passive — zero such modes — so the wall's central object is *undefined*, not violated. It binds only the rejected AeST/khronon modified-gravity limb (which fails Cassini at  $+6-14\sigma$ ). v2's “the covariant route is walled by strong coupling” applied that dynamical-aether wall to the passive-frame realization and is hereby corrected.

**(e) The constraint structure closes as a machine-verified Dirac system (v4).** The full Dirac–Bergmann analysis of the passive-frame + matter system in *curved* spacetime terminates cleanly. The primaries are  $p_\lambda \approx 0$  together with a *degenerate* Legendre map for the  $u$ -momenta (the velocity Hessian is proportional to the spatial  $k^2$ , so there is no invertible hyperbolic kinetic operator — the precise sense in which  $u$  carries no Cauchy data); the secondary is the constraint  $\chi_1: u \cdot u + 1 = 0$ ; and the  $u$ -equation of motion *algebraically* fixes the multiplier  $\lambda = -ps$  with **no tertiary tower**. The unit-norm pair  $(\chi_1, u \cdot \pi)$  is genuinely **second-class** — its Dirac block is  $[[0, 2(u \cdot u)], [-2(u \cdot u), 0]]$  with determinant  $4(u \cdot u)^2 = 4$  on-shell (machine-verified carrying the full metric  $g_{\mu\nu}$ ) — so a Dirac bracket exists and the frame-sector degree-of-freedom count is  $\frac{1}{2}[10-0-10]=0$ ; the only propagating modes are GR's two graviton polarizations. The crux that  $u$  sits *inside*  $\square_u$  is resolved honestly: the exact khronon variation gives a *nonzero* transverse quadratic form (the naive “identically zero, purely longitudinal” reading is false), but that form is  $S_n = (-1)^n k_\perp^2 \perp k_0^{2n}$ , whose only nonzero- $k_\perp$  root is  $k_0 = 0$  (a double root independent of spatial  $k$ ) — a transport-along- $u$  ODE with no wave-cone and zero group velocity, **not** a spatially propagating aether kinetic term. Two honest relabelings surfaced in verification, neither changing the count: hypersurface-orthogonality (twist=0) is the *khronon-frame postulate*, not an algorithm-generated secondary constraint on a generic unit 4-vector; and  $\pi_\mu \approx 0$  is a degenerate Legendre map, not a literal vanishing. The construction is thus mathematically complete **given the khronon-frame premise**.

**(f) The nonlocal operator  $K(\square_u)$  is rigorously defined (v4).** On a globally hyperbolic passive- $u$  background the directional operator  $\square_u = (u \cdot \nabla)^2$  is essentially self-adjoint, so  $K(\square_u) = \int K(\lambda) dE(\lambda)$  is fixed by the Borel functional calculus.  $K(z)$  is a genuine Herglotz–Nevanlinna function of  $z = \square_u/a_0^2$  — proven analytically via the uniformizer  $w = \sqrt{(1+4z)}$  ( $K = \sqrt{((w-1)/(w+1))}$  is a composition of upper-half-plane maps) and numerically (min Im  $K$  over the upper half-plane =  $+4.8 \times 10^{-6}$ , zero violations). The Herglotz–Nevanlinna representation theorem then supplies a *unique positive Borel measure* with closed-form density ( $\rho_A = (1 - \sqrt{(1-4|t|)})/(2\pi\sqrt{|t|})$  on  $-1/4 < t < 0$ ,  $\rho_B = 1/(2\pi\sqrt{|t|})$  on  $t < -1/4$ ), everywhere  $\geq 0$  (Källén–Lehmann), with  $\int d\mu/(1+t^2)$  finite; the representation reconstructs  $K$  to  $< 10^{-11}$ .  $K(\square_u)$  is bounded ( $\|K\| \leq 1$ , operator-monotone/Loewner), defined on all of  $L^2$ , and causal-retarded. The non-entire branch cut is the physical de Sitter–Unruh emission-threshold continuum, not an obstruction. One relabeling improves the physical reading:

$K(\square_u)$  is self-adjoint on the static  $z>0$  sector, while on the dynamical  $z<0$  (oscillatory) sector the spectrum lies on the cut and  $K(\square_u)$  is **normal** and causal-retarded (its positive anti-Hermitian part is the Källén–Lehmann-guaranteed dissipative response), not globally self-adjoint.

**What is open.** Nonlinear classical stability now follows from (e): the exact symbol  $S_n=(-1)^n k^2 \perp k_0^{(2n)}$  holds at every order, background gradients and curvature add strictly lower-derivative pieces that cannot flip the cone, so there is no nonlinear ghost or runaway and hyperbolicity is inherited from GR ( $K \rightarrow 1$  in the UV,  $K \rightarrow 0$  in the IR, both bounded). What remains open is **radiative** stability as a *computation*: the  $(\nabla u)^2$  aether-kinetic counterterm is symmetry-forbidden (every  $u$ -derivative in every vertex is longitudinal,  $u \cdot \nabla$ , so no pure-transverse piece can be generated —  $\alpha_1=\alpha_2=0$  is not a tuning that runs),  $a_0$  is IR-protected by the  $\sqrt{z}$  IR-analyticity (no  $z^0$  tadpole) plus khronon shift symmetry  $T \rightarrow T + \text{const}$ , and Källén–Lehmann positivity survives loops by the pump-free/KMS argument — but a full covariant background-field one-loop calculation on de Sitter (confirming no  $(\nabla u)^2$  counterterm, the  $a_0$ -vertex decoupling, and dressed positivity from a real loop integral) is not performed. A first-principles ghost-condensate profile law (§5) also remains open.

## 5. Gravitational lensing: viable but fine-tuned

In modified inertia the modification is to how massive bodies respond; **light is massless and has no inertia to modify**, so photons follow GR null geodesics of real mass — baryons plus the ghost-condensate dark sector. The observed weak-lensing RAR matches the dynamical RAR [Brouwer et al. 2021; Mistele & McGaugh 2024], which *requires* the condensate to supply exactly the MI dynamical excess,  $M_{\text{cond}}(r)=M_{\text{bar}}(v(y)-1)$ . Nothing in the framework forces this: the condensate amount is a free shift-charge integration constant ( $dI_0/da_0=0$ ), and generic clustering profiles miss the required shape by 0.12–0.55 dex — only a hand-placed profile matches. Worse, a fixed cosmic  $I_0$  gives a constant  $M_{\text{cond}}/M_{\text{bar}}$  while the MI excess runs as  $\sqrt{M_{\text{bar}}}$  across the galaxy population, so the tuning is *per-galaxy*. This is **not a tension** ( $M_{\text{cond}}=M_{\text{bar}}(v-1)$  is positive-density, monotonic, and realizable — the framework is not excluded) and **not a prediction** (it is a fit). Verdict: viable but fine-tuned, the location of generic two-sector dark theories. The genuinely distinctive modified-inertia signature lives in the *dynamical* sector (the MG-impossible non-adiabatic velocity-dispersion hysteresis), not in lensing.

## 6. Honest standing

Reasoning from the framework’s own passive-frame, MOND-in-matter premise: the worldline sector is written and verified; the modified-inertia realization evades Cassini; MOND does not force modified gravity; the mixed two-point propagator is causal, ghost-free (Källén–Lehmann), and principal-symbol-well-posed; the Einstein-aether strong-coupling wall is a mis-application that does not bind the framework (it binds only the rejected modified-gravity limb); and — new in v4 — the passive-frame **constraint structure closes as a machine-verified Dirac system with zero propagating frame modes**, the nonlocal operator  $K(\square_u)$  is **rigorously defined** by a unique positive Herglotz–Nevanlinna measure, and **nonlinear classical stability holds** (the no-wave-cone symbol to all orders). Gravitational lensing is viable but fine-tuned (per-galaxy), not solved and not a kill. The completion is therefore **not** the modified-gravity limb (Cassini-failing) and **not yet** a finished field theory — it is a well-posed worldline theory whose passive-frame covariantization has a closed constraint algebra (0 frame dof), a rigorously defined causal-retarded nonlocal operator, a causal ghost-free two-point sector, principal-symbol well-posedness, and nonlinear stability, with two honest open edges: a full covariant one-loop de Sitter computation certifying the (structurally protected) radiative stability, and a first-principles ghost-condensate profile law. The MOND sign stays postulated and  $a_0$ ’s value underived. No completeness or theory-of-everything claim is made.

## 7. Related work

The construction sits between the relativistic modified-gravity realizations of MOND — AeST [Skordis & Złośnik 2021], the Blanchet–Skordis khronon [2025], the nonlocal mimetic model of Deffayet & Woodard [2026] — and the modified-inertia program of Milgrom [1999, 2009]. The covariant frame sector is Einstein-aether [Jacobson & Mattingly 2001], its PPN structure [Foster & Jacobson 2006], its GW170817 constraint [Gong et al. 2018], and its strong-coupling corner [Blas, Pujolàs & Sibiryakov 2009]; the nonlocal well-posedness class is Källén–Lehmann/Herglotz–Nevanlinna (passive-causal), distinct from the entire-function class [Biswas et al.; Barnaby & Kamran]. The doubled-worldline formalism is Galley [2013]; the lensing constraint is Brouwer et al. [2021] and Mistele & McGaugh [2024].

## 8. Reproducibility

Committed scripts (repository `real_research/reviews/`): the worldline sector and Cassini evasion (`cassini_mi_evasion_2026/`); the covariant kinetic wins (`mi_kinetic_completion_2026/`); the mixed-propagator causality + Källén–Lehmann positivity (`mi_propagator_2026/`); the framework-first correction of the strong-coupling wall (`mi_strongcoupling_framework_2026/`); the all-orders Cauchy checks (`mi_cauchy_wellposed_2026/`); the lensing fine-tuning analysis (`mi_lensing_2026/`); and — new in v4 — the closed Dirac constraint analysis, the no-wave-cone crux symbol, the Herglotz–Nevanlinna operator definition, and the nonlinear/radiative stability analysis (`mi_formal_completion_2026/`: `A10_dirac_block.py`, `A6b_higher_n.py`, `constraint_structure.py`, `crux_variation_check.py`, `operator_definition.py`, `stability.py`, `transverse_mode_analysis.py`). Sign-wall inputs: `ACTIVE_KERNEL_SIGNTHEOREM_2026-06.md`, `FOURTH_ORDER_SIGN_WALL_2026-07.md`; tri-chotomy: `COVARIANT_MI_COMPLETION_2026-06.md`.

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14. Companion (this program): sign wall, Zenodo 10.5281/zenodo.21184373; covariant-MI quadrichotomy, 10.5281/zenodo.21148494; written MI action v1/v2/v3, 10.5281/zenodo.21253645 / 21260830 / 21263846.

\*Both  $a_0$  footings throughout. Each claim labeled derived / postulated / open. New in v4: the passive-frame constraint structure closes as a machine-verified Dirac system (0 propagating frame dof), the nonlocal

operator  $K(\square_u)$  is rigorously defined by a unique positive Herglotz–Nevanlinna measure, and nonlinear classical stability holds; the one open formal edge is a covariant one-loop de Sitter computation certifying the (structurally protected) radiative stability. The MOND sign is a postulate; the Einstein-aether wall is a mis-application; lensing is viable-but-tuned. No completeness or theory-of-everything claim.\*